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Multigram-scale total synthesis of (±)-ambreinolide by a diastereoselective polyene cyclization

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ABSTRACT

Ambreinolide is a natural terpenoid with great value for perfume industry and natural product synthesis. Herein we report a novel total synthesis of ambreinolide on multigram-scale that employs a regio- and diastereoselective, high yielding, proton-initiated polyene cyclization using a catalyst easily generated *in situ*. Molecular structures were unambiguously confirmed by X-ray crystallography.

Ambergris ("grey amber") is a coprolite from the sperm whale (*Physeter macrocephalus*) digestive tract, probably arising from gut irritation by indigestible squid beaks. Its unique scent led to its use as a flavoring agent, incense, and as a perfume base since antiquity. It still is one of the best fixatives known for perfume production to date.[1] However, the trade of ambergris jetsam is banned by many countries,[2] and a shrinking sperm whale population and increasing pollution make ambergris extremely rare.[3] Therefore, it is typically substituted by synthetic alternatives.

As a component of the ambergris' odorless fraction, ambreinolide (**1**, Fig. 1), contributes to its extraordinary fixative properties.[3] It is believed to arise from ambreine (**2**), the main constituent of ambergris, by slow autoxidative degradation in the open sea.[4] Furthermore, it was found in several plants used for medical applications.[5–11] Several syntheses feature ambreinolide (**1**) as an intermediate or as a starting material.[12–20] Its terpene framework is typical for the drimane sesquiterpenoids.[21]

Previously reported syntheses of ambreinolide (**1**) mainly rely on transformations of other complex natural products which are not always easy to access.[12,14,15,22–26] An acid-mediated polyene cyclization of the far more simple farnesylacetic acid was reported early on by Stork, but gave less than 3% yield of racemic ambreinolide (**1**).[27] A more productive modification of this procedure uses 2-nitropropane as a non-benign solvent.[15,16] However, in our hands, this procedure was difficult to reproduce and generated numerous byproducts which were difficult to separate, resulting in low overall yields.

In order to develop a novel, scalable total synthesis of ambreinolide (**1**) from a polyene precursor, its structure was retrosynthetically simplified to enone **3** (Scheme 1), which may be accessed from phenolic derivatives **4** by reductive dearomatization. This terpenoid could rapidly arise from suitably protected dienes **5** by a proton-initiated, diastereoselective polycyclization reaction that is terminated by the electron-rich arene nucleus. Diene **5** was planned to be synthesized from easily available geranyl acetate (**6**) and a benzylic Grignard (**7**). Key to this plan was the development of a scalable polyene cyclization.[28]

Aiming to investigate and develop such a synthesis, the phenolic polyene **9** was prepared from geranyl acetate (**6**) by employing S_N2-selective substitution with a cuprate derived from Grignard reagent **7** (Scheme 2).[29] Subsequent demethylation of anisole **5a** afforded the phenol **5b** in very good yields.

After unsuccessful attempts to promote polyene cyclizations of this precursor with simple reagents, we turned our attention to Yamamoto's Lewis-acid assisted chiral Brønsted acids (LBAs)[30] and related reagents for stereoselective protonation.[31] These reagents are formed by coordination of a Lewis acid to a weak Brønsted acid, in order to increase its acidity, and to confine the putative protonation vector.[32] Enantioselective polyene cyclizations with LBAs of type **8**, formed from chiral BINOL derivatives with tin tetrachloride *in situ*, have been reported.[28] Owing to the preferred protonation of the terminal alkene, comparable to terpene cyclases, this method is considered biomimetic.[33–37]

We employed monobenzylated *S*-BINOL **10** as the Brønsted acid for studying the intended polyene cyclization and investigated different

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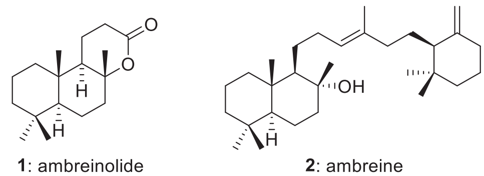
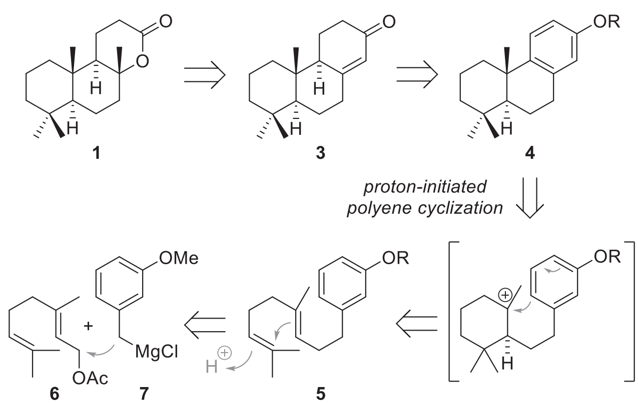
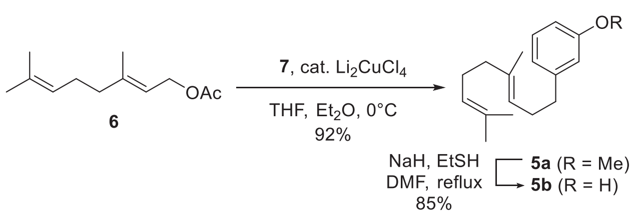


Fig. 1. Structures of ambreinolide (**1**) and ambreine (**2**).



Scheme 1. Retrosynthetic planning for the synthesis of ambreinolide (**1**).



Scheme 2. Synthesis of phenol-terminated diene **9**.

protecting groups for phenol **5b** (Table 1, entries 1–7) for achieving the desired regioselectivity and for obtaining proper reactivity.

Weakly electron-poor acylated phenols were found promising with respect to regio- and diastereoselectivity (entries 1–3). In these cases, additional exposure to BF_3•OEt_2 was required to complete bicyclization to tricycles **4c-f** from mixtures with monocyclized alkenes **9**. Nevertheless, the products were isolated as the desired single diastereo- and regioisomers in very good yields. The pivaloyl protection provided best yield and enantioselectivity (entry 3). Introduction of a phenylcarbamate group instead led to reduced enantiomeric excess (entry 4). Less electron-withdrawing protecting groups like dimethylcarbamate and TBDPS (entries 5–6) resulted in inseparable product mixtures. The strongly electron-withdrawing triflate disabled the second cyclization step (entry 7).

Although the pivaloyl-protected phenol **5e** was most productive, we observed only moderate enantioselectivity (62% e.e.). Therefore, a series of different BINOL derivatives **11**–**21** were tested as alternative reagents. Unfortunately, all modifications led to decreased enantioselectivity or discouraging loss of activity (entries 8–18). Changing the solvent or employing less active LBAs at higher temperature remained unsuccessful. The application of an antimony pentachloride based LBA[38] resulted in fully racemic product in this case (data not shown).

A mechanistic rationale delineated from Yamamoto's studies on LBA-induced asymmetric polyene cyclization is depicted in Scheme 3. LBA complexes **8** likely chelate $SnCl_4$ in an equatorial-equatorial fashion. The only moderately face selective protonation traverses putative transition state **I** in which steric repulsion between the BINOL moiety and the polyene substrate is minimized, and the n–π* interaction between an oxygen lone pair and the terminal double bond is facilitated.[33] Initial protonation could lead to a concerted transition state **II** that produces decaline **4e** directly by bicyclization. Alternatively, the polyene cyclization may proceed stepwise by involving the aryl-shielded intermediate **III**. This would give first the alkene isomers **9**. After a second protonation carbocationic intermediate **IV** would transform to decaline **4e** with high *trans*-selectivity.[35] The stepwise mechanism may be predominant considering the initial formation of alkenes **9** in large amounts (Table 1), with some bicyclization occurring in this application.

To our delight, we found the simple LBA (±)-BINOL·$SnCl_4$ *used in a sub-stochiometric amount* to preserve full regio- and diastereoselectivity, and to give high yields for the desired bicyclization (Scheme 4) after addition of BF_3•OEt_2. Unfortunately, lowering the amount of catalyst or using BF_3•OEt_2 or $SnCl_4$ alone produced product mixtures. Further optimization led to a one-pot procedure without the necessity to isolate intermediates, while BINOL can easily be recovered and reused.[38] Removal of the pivaloyl protecting group from bicyclization product **4e** provided highly crystalline phenol **4b** in very good yield on decagram scale (Scheme 4). X-ray single-crystal structure analysis confirmed the desired *trans* configuration.

Methylation of phenol **4b** gave anisole **4a** quantitatively (Scheme 5). Birch reduction and acidic hydrolysis of the resulting enol ether afforded enone **3** selectively in very good yield.

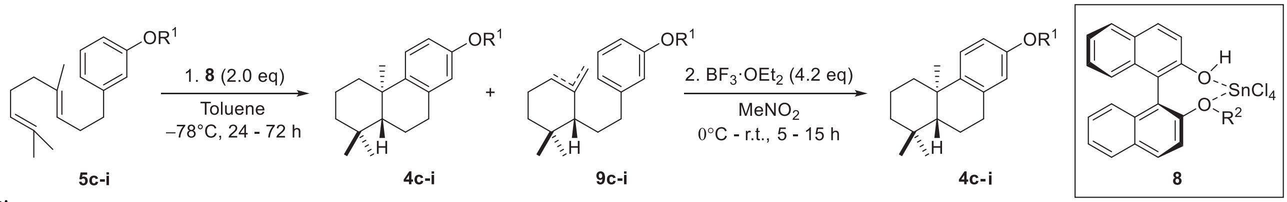
Treatment of enone **3** with *m*CPBA induced Baeyer-Villiger oxidation and subsequent enol epoxidation,[24] to cleanly yield epoxy lactone **22** (Scheme 6). Wolff-Kishner-type conditions led to base-induced rearrangement and formation of intermediate hydrazone **23**, which converted to acid **24** with heating.[24,39] Upon exposure to sulfuric acid in acetic acid, carboxylic acid **24** underwent a cationic cyclization to selectively give ambreinolide **(±)-1** in an overall yield of 65% from enone **3**. X-ray single-crystal structure analysis confirmed the desired structure of the obtained product (Fig. 2). We synthesized 7.7 g of ambreinolide **(±)-1** in a single lab batch, demonstrating the scalability and robustness of this total synthesis route.

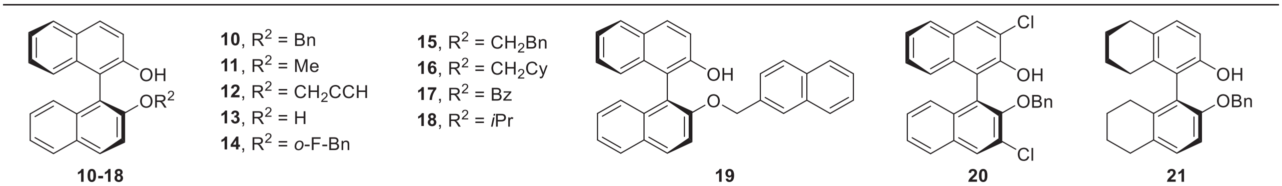
In summary, we have developed a novel total synthesis of racemic ambreinolide, providing convenient access to **(±)-1** on multigram-scale. This material is not only highly valuable *per se*, but also an attractive starting material for the synthesis of structurally related natural products. Key was the development of a regio- and diastereoselective, high-yielding polyene cyclization induced by the LBA (±)-BINOL·$SnCl_4$, of which the BINOL component can be easily recycled. These results will contribute to future terpene total synthesis projects and increase accessibility of rare materials from nature for human use.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

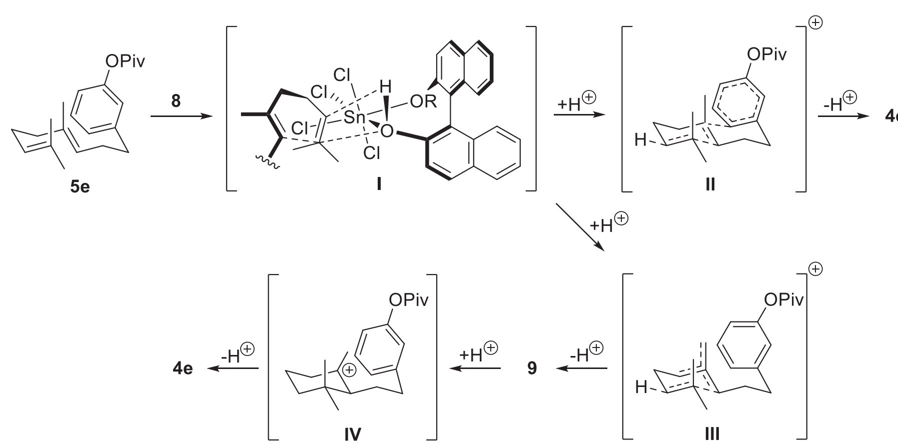
Table 1
Optimization of enantioselective polyene cyclization of dienes **5** with reagents **8** from 10–21.



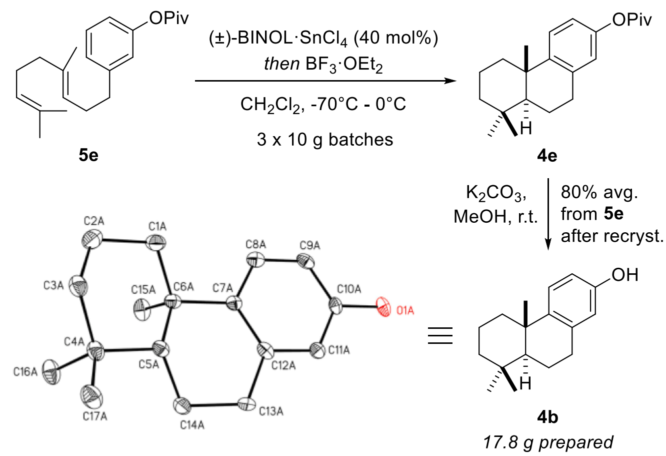


Entry	Cmp.	R_1 =	BINOL	Step 1[a]		Step 2	
				Ratio (**4:9**)[b]	Yield [%][c]	Yield [%][c]	ee [%][d]
1	**5c**	Ac	**10**	3.1:6.9	86	93 (**4c**)	52
2	**5d**	Bz	**10**	2.0:8.0	85	94 (**4d**)	58
3	***5e***	***Piv***	***10***	***2.4:7.6***	***92***	***92*** (4e)	***62***
4	**5f**	C(O)NHPh	**10**	3.4:6.6	90	92 (**4f**)	52
5	**5g**	C(O)NMe_2	**10**	0.6:9.4	90	product mixture	
6	**5h**	TBDPS	**10**	1.1:8.9	87	product mixture	
7	**5i**	Tf	**10**	0.0:1.0	84	product mixture	
8	**5e**	Piv	**11**	2.8:7.2	96	93 (**4e**)	44
9	**5e**	Piv	**12**	2.8:7.2	94	92 (**4e**)	42
10	**5e**	Piv	**13**	2.8:7.2	94	91 (**4e**)	30
11	**5e**	Piv	**14**	1.3:8.7	95	92 (**4e**)	58
12	**5e**	Piv	**15**	1.0:9.0	92	91 (**4e**)	48
13	**5e**	Piv	**16**	low conversion		–	
14	**5e**	Piv	**17**	no conversion		–	
15	**5e**	Piv	**18**	no conversion		–	
16	**5e**	Piv	**19**	1.4:8.6	94	91 (**4e**)	54
17	**5e**	Piv	**20**	3.9:6.1	90	90 (**4e**)	54
18	**5e**	Piv	**21**	no conversion		–	

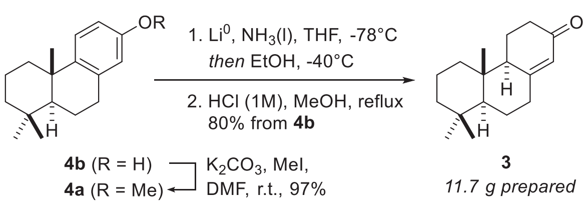
[a] Reactions were conducted at 0.3 mmol scale.
[b] Determined by ^{1}H NMR analysis.
[c] Isolated yields of products **4c**-**4f**.
[d] Determined by HPLC analysis.



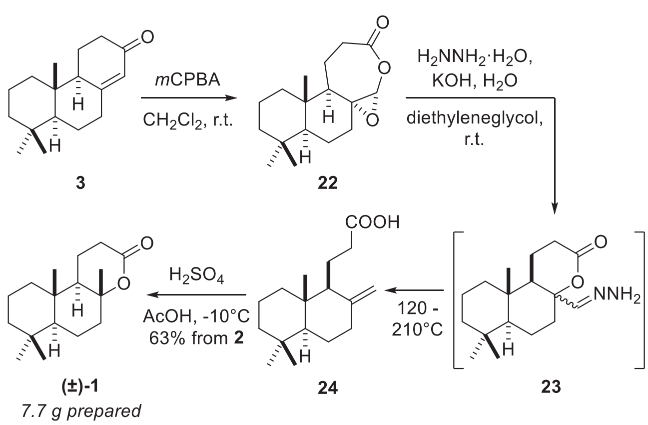
Scheme 3. Proposed mechanism for LBA-induced asymmetric polyene cyclization.



Scheme 4. Optimized synthesis of phenol **4b**. Molecular structure and atom numbering of phenol **4b** in the solid state shown. The ellipsoids represent a probability of 30 %. Hydrogen atoms are omitted for clarity.



Scheme 5. Synthesis of enone **3**.



Scheme 6. Synthesis of (±)-ambreinolide **1**.

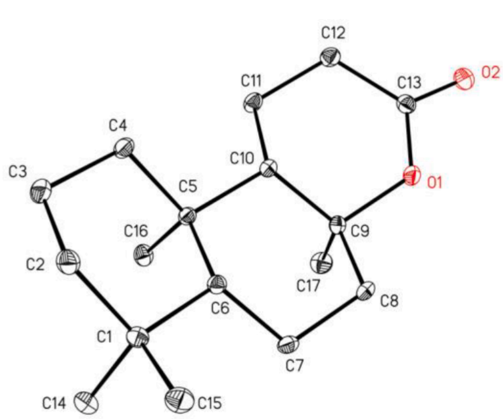


Fig. 2. Molecular structure and atom numbering scheme of (±)-ambreinolide (**1**). The ellipsoids represent a probability of 30 %. Hydrogen atoms are omitted for clarity.

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Appendix A. Supplementary data

Crystallographic data (excluding structure factors) has been deposited with the Cambridge Crystallographic Data Centre as supplementary publication CCDC-2162308 for **1**, and CCDC-2162309 for **4b**. Copies of these data can be obtained free of charge on application to CCDC, 12 Union Road, Cambridge CB2 1EZ, UK [E-mail: deposit@ccdc.cam.ac.uk]. Supplementary data to this article can be found online at https://doi.org/10.1016/j.bmcl.2022.128845.

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